S1. Personal annual expenditure

Source: Annual expenditure!A1:G22; full baseline, personal shares already allocated.

Workbook: Labels.xlsx | Provisional assessment | 2026-11-07

Row	Expenditure item	Monthly AUD	Months	Annual AUD	Basis
2	Shared apartment rent	1,300.000	12.000	15,600.000	Monthly × 12
3	Groceries and non-alcoholic drinks	400.000	12.000	4,800.000	Monthly × 12
4	Restaurants, cafes and takeaway	200.000	12.000	2,400.000	Monthly × 12
5	Electricity	40.000	12.000	480.000	Monthly × 12
6	Household gas	10.000	12.000	120.000	Monthly × 12
7	Water	10.000	12.000	120.000	Monthly × 12
8	Home internet	30.000	12.000	360.000	Monthly × 12
9	Mobile phone service	30.000	12.000	360.000	Monthly × 12
10	Public transport	150.000	12.000	1,800.000	Monthly × 12
11	Taxi and rideshare	20.000	12.000	240.000	Monthly × 12
12	Clothing and footwear	50.000	12.000	600.000	Monthly × 12
13	Small household goods and cleaning products	30.000	12.000	360.000	Monthly × 12
14	Toiletries and personal-care products	40.000	12.000	480.000	Monthly × 12
15	Medical, dental and pharmaceutical expenses	50.000	12.000	600.000	Monthly × 12
16	Recreation and entertainment services	50.000	12.000	600.000	Monthly × 12
17	Books and learning materials	30.000	12.000	360.000	Monthly × 12
18	Contents insurance	20.000	12.000	240.000	Monthly × 12
19	Hairdressing	20.000	12.000	240.000	Monthly × 12
20	Flights and accommodation	0.000	12.000	0.000	Monthly × 12
21	Vehicle purchase, fuel and maintenance	0.000	12.000	0.000	Monthly × 12
	TOTAL			29,760.000	AUD/person/year

Hypothetical Redfern adult; two-adult shared apartment; no private car. Rent AUD 300/week = 15,600/year. Shared costs are personal shares, not household totals.

S2. Mixed-expenditure allocation

Source: Expenditure split!A1:I41. All 40 allocation rows shown; percentages are explicit scenario assumptions.

Workbook: Labels.xlsx | Provisional assessment | 2026-11-07

Source row	Split component	Sector ID	Share	Annual AUD	IO sector
2.000	Rent maps to dwelling services, not building construction or real-estate agent fees.	292.000	100%	15,600.000	Ownership of dwellings
3.000	Fresh meat	75.000	15%	720.000	Fresh meat
3.000	Poultry	77.000	10%	480.000	Poultry, slaughtered
3.000	Milk	79.000	10%	480.000	Treated milk
3.000	Cheese	80.000	5%	240.000	Cheese
3.000	Vegetables	15.000	15%	720.000	Vegetables
3.000	Fruit	16.000	10%	480.000	Fruit
3.000	Bread	96.000	10%	480.000	Bread and bread rolls
3.000	Pasta	94.000	5%	240.000	Pasta
3.000	Rice products	86.000	5%	240.000	Rice products
3.000	Other food	98.000	10%	480.000	Food products
3.000	Soft drinks	105.000	5%	240.000	Soft drinks
4.000	Purchased meals; no separate ingredient mapping.	259.000	100%	2,400.000	Hotels, clubs, restaurants and cafes
5.000	Electricity supply	241.000	100%	480.000	Electricity supply
6.000	Gas supply	242.000	100%	120.000	Gas supply
7.000	Water/sewerage services	243.000	100%	120.000	Water supply; sewerage and drainage services
8.000	Domestic home internet service	282.000	100%	360.000	Domestic telecommunication services
9.000	Mobile service only	282.000	100%	360.000	Domestic telecommunication services
10.000	Passenger rail	265.000	60%	1,080.000	Railway passenger transport services
10.000	Bus/tram/light rail	262.000	40%	720.000	Bus and tramway

S2. Mixed-expenditure allocation

Source: Expenditure split!A1:I41. All 40 allocation rows shown; percentages are explicit scenario assumptions.

Workbook: Labels.xlsx | Provisional assessment | 2026-11-07

Source row	Split component	Sector ID	Share	Annual AUD	IO sector
11.000	Taxi/rideshare proxy	263.000	100%	240.000	Taxi and hire car
12.000	Clothing	118.000	80%	480.000	Clothing
12.000	Footwear	119.000	20%	120.000	Footwear
13.000	Cleaning products	149.000	50%	180.000	Soap and other detergents
13.000	Household paper products	130.000	25%	90.000	Paper products
13.000	Small plastic household goods	160.000	25%	90.000	Plastic products
14.000	Toiletry products	150.000	75%	360.000	Cosmetics and toiletry preparations
14.000	Soap/detergents for personal use	149.000	25%	120.000	Soap and other detergents
15.000	GP/dental/optometry services	322.000	50%	300.000	GPs, dentists, optometrists, ambulance
15.000	Pharmaceuticals	148.000	50%	300.000	Pharmaceutical goods for human use
16.000	Cinema	326.000	40%	240.000	Motion pictures
16.000	Pay-TV/streaming proxy; imperfect sector match	279.000	30%	180.000	Overseas communication services and Pay TV
16.000	Sport/recreation services	337.000	30%	180.000	Sport and recreation services (incl horse and dog racing, sports grounds, services)
17.000	Books	137.000	80%	288.000	Books, maps, magazines
17.000	Stationery	131.000	20%	72.000	Printing and stationery
18.000	Contents insurance: premium used as provisional service-spending proxy; insurance valuation unresolved.	285.000	100%	240.000	Insurance
19.000	Hairdressing personal services	338.000	100%	240.000	Personal services
20.000	Zero baseline; placeholder allocation only. Add accommodation split if future travel scenario changes.	269.000	50%	0.000	Air and space transport
20.000	Zero baseline accommodation placeholder	260.000	50%	0.000	Accommodation
21.000	Zero baseline; fuel placeholder only. Rebuild vehicle purchase/maintenance splits if profile changes.	139.000	100%	0.000	Petrol and diesel

Check source allocation shares total 100%. The full purchaser-price mapping remains provisional: taxes, margins and 2018 price conversion are not resolved.

S3. 345-sector final demand

Source: IO_345 demand!A1:G347. Nonzero baseline sectors shown; zero-demand sectors remain in the full 345-row vector.

Workbook: Labels.xlsx | Provisional assessment | 2026-11-07

Sector ID	Sector name	Annual AUD	y (AUD million)	x (AUD million)	Q (kt CO2-e)	
15	Vegetables		720.000	0.000720	5,215.800	1,177.800
16	Fruit		480.000	0.000480	4,085.100	936.390
75	Fresh meat		720.000	0.000720	14,177.000	1,289.100
77	Poultry, slaughtered		480.000	0.000480	4,463.700	972.620
79	Treated milk		480.000	0.000480	105.610	266.390
80	Cheese		240.000	0.000240	2,703.500	301.560
86	Rice products		240.000	0.000240	463.180	41.096
94	Pasta		240.000	0.000240	704.600	67.935
96	Bread and bread rolls		480.000	0.000480	3,877.700	310.560
98	Food products		480.000	0.000480	7,564.100	796.660
105	Soft drinks		240.000	0.000240	5,854.000	629.070
118	Clothing		480.000	0.000480	3,227.200	1,363.900
119	Footwear		120.000	0.000120	933.030	437.140
130	Paper products		90.000	0.000090	2,830.000	346.210
131	Printing and stationery		72.000	0.000072	3,628.400	308.880
137	Books, maps, magazines		288.000	0.000288	2,536.700	46.136
148	Pharmaceutical goods for human use		300.000	0.000300	9,899.600	2,035.100
149	Soap and other detergents		300.000	0.000300	359.570	71.377
150	Cosmetics and toiletry preparations		360.000	0.000360	857.680	120.920
160	Plastic products		90.000	0.000090	10,427.000	1,135.900
241	Electricity supply		480.000	0.000480	43,542.000	60,073.000
242	Gas supply		120.000	0.000120	3,772.900	0.020
243	Water supply; sewerage and drainage services		120.000	0.000120	29,893.000	14,851.000
259	Hotels, clubs, restaurants and cafes		2,400.000	0.002400	61,016.000	195.050
262	Bus and tramway		720.000	0.000720	7,742.000	1,647.400
263	Taxi and hire car		240.000	0.000240	5,771.900	2,386.500
265	Railway passenger transport services		1,080.000	0.001080	7,549.000	2,395.800
279	Overseas communication services and Pay TV		180.000	0.000180	5.862	0.000
282	Domestic telecommunication services		720.000	0.000720	44,642.000	152.140
285	Insurance		240.000	0.000240	40,356.000	106.800
292	Ownership of dwellings		15,600.000	0.015600	250,390.000	35.241
322	GPs, dentists, optometrists, ambulance		300.000	0.000300	38,991.000	147.190
326	Motion pictures		240.000	0.000240	6,381.500	21.279
337	Sport and recreation services (incl horse and dog racing, sports grounds, services)		180.000	0.000180	19,177.000	73.222
338	Personal services		240.000	0.000240	46,294.000	173.430
	TOTAL		29,760.000	0.029760		

y = annual AUD / 1,000,000. T and x are AUD million; Q is kt CO2-e. Matrix/vector order follows row labels exactly.

S4. Teacher-method matrix construction

Sources: Xhat, Xhat inverse, A coefficients and L matrix. Exact array formulas and top-left matrix excerpts.

Workbook: Labels.xlsx | Provisional assessment | 2026-11-07

Step	Source cell / array	Formula / meaning
Xhat	'$\mathbf{X}$̂'!A1:MG345	Formula: =_xlfn.MUNIT(345)*Output!A1:A345
Xhat inverse	'$\mathbf{X}$̂ inverse'!A1:MG345	Formula: =MINVERSE('$\mathbf{X}$̂'!A1:MG345)
A	'A coefficients'!B2:MH346	Formula: A_ij = T_ij / x_j; equivalent to MMULT(T, Xhat inverse)
L	'L matrix'!B2:MH346	Formula: =MINVERSE(_xlfn.MUNIT(345)-'A coefficients'!B2:MH346)

$\mathbf{X}$̂ (5 × 5 excerpt)

Row 1	3319.800000	0.000000	0.000000	0.000000	0.000000
Row 2	0.000000	2264.600000	0.000000	0.000000	0.000000
Row 3	0.000000	0.000000	1704.800000	0.000000	0.000000
Row 4	0.000000	0.000000	0.000000	7567.400000	0.000000
Row 5	0.000000	0.000000	0.000000	0.000000	2255.400000

$\mathbf{X}$̂ inverse (5 × 5 excerpt)

Row 1	0.000301	0.000000	0.000000	0.000000	0.000000
Row 2	0.000000	0.000442	0.000000	0.000000	0.000000
Row 3	0.000000	0.000000	0.000587	0.000000	0.000000
Row 4	0.000000	0.000000	0.000000	0.000132	0.000000
Row 5	0.000000	0.000000	0.000000	0.000000	0.000443

A coefficients (5 × 5 excerpt)

Row 1	0.002492	0.001924	0.002110	0.002091	0.002125
Row 2	0.000451	0.000349	0.000458	0.000379	0.000385
Row 3	0.021384	0.016336	0.017516	0.017725	0.018136
Row 4	0.051639	0.039510	0.039648	0.042814	0.043790
Row 5	0.006924	0.005416	0.005737	0.005756	0.005867

L matrix (5 × 5 excerpt)

Row 1	1.003606	0.002783	0.003105	0.003021	0.003072
Row 2	0.001158	1.000898	0.001040	0.000974	0.000988
Row 3	0.024779	0.018949	1.020354	0.020555	0.021022
Row 4	0.058693	0.044937	0.045430	1.048687	0.049783
Row 5	0.008322	0.006500	0.006918	0.006925	1.007057

S5. q, m and footprint calculation

Sources: q matrix!B2:MH2, m matrix!B2:MH2, Footprint sectors and Footprint summary.

Workbook: Labels.xlsx | Provisional assessment | 2026-11-07

Step	Source	Exact formula / units
q	'q matrix'!B2:MH2	Formula: =MMULT(Satellite!A1:MG1,'X^ inverse'!A1:MG345)
m	'm matrix'!B2:MH2	Formula: =MMULT('q matrix'!B2:MH2,'L matrix'!B2:MH346)
Contribution	'Footprint sectors'!G2:G346	Formula: G_i = m_i * y_i * 1000 (tonnes CO2-e)
Induced output	'Footprint sectors'!I2	Formula: =MMULT('L matrix'!B2:MH2,$D$2:$D$346)
Total	'Footprint summary'!B3	Formula: =SUM('Footprint sectors'!G2:G346)

Sector ID	Purchase sector	q (kt/AUD m)	m (kt/AUD m)	Annual AUD	t CO2-e/year
75	Fresh meat	0.091	2.376	720.000	1.711
79	Treated milk	2.522	2.727	480.000	1.309
241	Electricity supply	1.380	2.456	480.000	1.179
77	Poultry, slaughtered	0.218	2.075	480.000	0.996
259	Hotels, clubs, restaurants and cafes	0.003	0.209	2,400.000	0.502
265	Railway passenger transport services	0.317	0.455	1,080.000	0.491
292	Ownership of dwellings	0.000	0.018	15,600.000	0.278
	TOTAL t CO2-e/person/year	9.543			

Multiplication uses compatible dimensions: q (1×345), L (345×345), y (345×1). qLy yields kt; multiply by 1,000 for tonnes.

S6. Validation and reconciliation

Sources: Validation and Footprint checks. Live reconciliation formulas; independent numerical residuals documented separately.

Workbook: Labels.xlsx | Provisional assessment | 2026-11-07

Check	Description	Value	Expected	
Validation	Raw annual total	29,760.000	29,760.000	0.000 Difference = 0
Validation	Split annual total	29,760.000	29,760.000	0.000 Difference = 0
Validation	345-sector AUD total	29,760.000	29,760.000	0.000 Difference = 0
Validation	Provisional y million AUD	0.030	0.030	0.000 Difference = 0
Validation	Sector count	345.000	345.000	0.000 Difference = 0

Source	Description	Value	Acceptance
Footprint checks	Minimum total output x	0.080	> 0
Footprint checks	max abs((I-A)L-I)	0.000	< 1e-10
Footprint checks	max abs((I-A)Ly-y)	0.000	< 1e-10
Footprint checks	sum consumption minus qLy (t)	0.000	absolute value < 1e-9
Footprint checks	sum emitting sectors minus qLy (t)	0.000	absolute value < 1e-9
Footprint checks	Condition number of I-A	5.778	Diagnostic; no universal pass threshold
Footprint checks	Consumption total minus emitting-sector total (t)	0.000	absolute value < 1e-9

Algebraic checks establish numerical consistency, not completeness of emissions boundaries or price/valuation comparability. See evidence/validation.json for the current independent audit.

S7. Provisional footprint and sector hotspots

Source: Footprint summary!A2:D22. Consumption categories driving emissions; not emitting-sector locations.

Workbook: Labels.xlsx | Provisional assessment | 2026-11-07

Rank	Consumption category	t CO2-e/year	Share
1.000	Fresh meat	1.711	17.9%
2.000	Treated milk	1.309	13.7%
3.000	Electricity supply	1.179	12.4%
4.000	Poultry, slaughtered	0.996	10.4%
5.000	Hotels, clubs, restaurants and cafes	0.502	5.3%
6.000	Railway passenger transport services	0.491	5.1%
7.000	Vegetables	0.360	3.8%
8.000	Ownership of dwellings	0.278	2.9%
9.000	Clothing	0.265	2.8%
10.000	Cheese	0.256	2.7%
	Total	9.543	

PRELIMINARY: nominal expenditure × 2018 intensities. Price basis, purchaser/basic-price valuation, insurance and direct household combustion remain unresolved. Australian/international comparisons are not yet available.